



Twenty-Five Things You Didn't Know You Could Do with R

This blog post contains code samples for my keynote at [Cascadia R Conference 2025](#). In addition, I've added resources for those interested in learning more. I'll also post the recording of my talk when it is available.

Slides

Twenty-Five Things You Didn't Know You Could Do with R

Twenty-Five Things You Didn't Know You Could Do with R

Access Data Automagically

Pull in Data Directly from Google Sheets

Sample Code

```
library(google Sheets4)

survey_data <-
  read_sheet(YOURSHEETURLHERE)
```



Resources

- I talk about how to use {googlesheets4} in [Chapter 11 of my book R for the Rest of Us: A Statistics-Free Introduction](#).
- [Check out the {googlesheets4} package docs](#).

Pull in Data Directly from Qualtrics

Sample Code

```
library(qualtRics)

survey_data <-
  fetch_survey(surveyID = "YOURSURVEYIDHERE")
```

Resources

- [Check out the {qualtRics} package](#)

Pull in Data Directly from the Census Bureau

Sample Code

```
library(tidycensus)

get_acs(
  state = "OR",
  geography = "place",
  geometry = TRUE,
  variables = "B01003_001"
)
```

Resources

- Great places to learn more are the [tidycensus docs](#) and [Kyle Walker's book about analyzing US census data](#)

Work with APIs to Access Data

Sample Code

This code doesn't work (you won't have an API key to make it work), but should give you a sense of how {httr2} works.

```
library(httr2)

fathom_api_key <- Sys.getenv("FATHOM_API_KEY")

request("https://api.usefathom.com/v1/aggregations") |>
  req_url_query(
    entity = "pageview",
    aggregates = "visits,uniques,pageviews",
    sort_by = "visits:desc"
  ) |>
  req_headers(
    Authorization = str_glue("Bearer {fathom_api_key}")
  ) |>
  req_perform()
```

Resources

- Check out [{httr2} package docs](#) and [this presentation about getting data out of Web APIs by Ted Laderas](#)

Scrape Data

```
library(rvest)
library(tidyverse)

read_html("https://en.wikipedia.org/wiki/List_of_FIFA_World_Cup_finals")
  html_elements("table") |>
  pluck(4) |>
  html_table() |>
  select(-Ref.)
```

Sample Code

Resources

- See [{rvest} package docs](#)

Efficiently Make Beautiful Data Viz

Make Your Own Theme

Sample Code

Here is the full code for my theme:

```
theme_dk <- function(base_family = "Inter Tight", base_size = 14) {
  theme_dk <-
    ggplot2::theme_minimal(
      base_size = base_size,
      base_family = base_family
    ) +
    ggplot2::theme(
      panel.grid.minor = ggplot2::element_blank(),
      panel.grid.major = ggplot2::element_line(
        color = "grey90",
        linewidth = 0.5,
        linetype = "dashed"
      ),

```

```

axis.ticks = ggplot2::element_blank(),
axis.text = ggplot2::element_text(
  color = "grey50",
  size = ggplot2::rel(0.8)
),
axis.title = ggplot2::element_blank(),
plot.title.position = "plot",
plot.title = ggplot2::element_text(
  face = "bold",
  size = ggplot2::rel(1.5)
),
plot.subtitle = ggplot2::element_text(
  color = "grey40",
  size = ggplot2::rel(1.1)
),
plot.caption = ggplot2::element_text(
  color = "grey50",
  margin = ggplot2::margin(t = 20)
),
plot.margin = ggplot2::margin(10, 10, 10, 10),
strip.text = ggplot2::element_text(
  color = "grey40",
  size = ggplot2::rel(0.9)
),
panel.spacing = ggplot2::unit(2, "lines")
)

theme_dk
}

```

Resources

Check out the *Customize Your Theme* lesson video from my *Going Deeper with R* course.

```

3 theme(axis.text.y = element_text(),
4       legend.position = "none",
5       plot.title = element_markdown(size = 20),
6       plot.title.position = "plot")

```



In 2007, **Dream island** had the most penguins of any island



Transcript

Click on the transcript to go to that point in the video. Please note that transcripts are auto generated and may contain minor inaccuracies.

0:03 If you've used themes in Ggplot, you've seen how they can change the whole look and feel of your plot with just a single line of code. The really cool thing about Ggplot is that you don't just have to rely on themes made by others. You can actually make your own theme so you can have all of your preferences baked in. And then you can apply them

I also talk about it in [Chapter 3 of my book R for the Rest of Us: A Statistics-Free Introduction](#).

Use Your Theme Everywhere

Sample Code

```
---  
title: "Penguins Report"  
---  
  
```${r}  
theme_dk <- function(base_family = "Inter Tight", base_size = 14) {}
```${r}  
  
```${r}  
theme_set(theme_dk())
```

```
```\n\n```\n{r}\npenguins_bar_chart\n```\n
```

Resources

Learn more about modifying themes in [theme\(\) function documentation](#)

Make Your Text Consistent with Your Theme

Sample Code

```
---\ntitle: "Penguins Report"\n---\n\n```\n{r}\ntheme_dk <- function()\n```\n\n```\n{r}\ntheme_set(theme_dk())\n\nupdate_geom_defaults(geom = "text", aes(family = "IBM Plex Mono"))\n```\n\n```\n{r}\npenguins_bar_chart +\n  geom_text(\n    aes(label = avg_bill_length_formatted),\n    vjust = 1.5,\n    color = "white",\n    size = 4\n  )\n```\n
```

Resources

- See the [update_geom_defaults\(\) documentation](#)

Make Maps

Make Maps with ggplot

Sample Code

```
library(tidycensus)

median_income_by_county <-
  get_acs(
    geography = "county",
    variables = c(median_income = "B19013_001"),
    geometry = TRUE
  )
```

Resources

- [Chapter 4 of my book](#) covers making maps with ggplot, and you can learn more in the [Mapping with R course](#)

Do Geospatial Analysis

Sample Code

```
library(sf)
library(tidyverse)
library(mapgl)
```

```

portland_libraries <- read_st("https://raw.githubusercontent.com/rfortherestofus/portland_libraries/master/portland_libraries.csv")

portland_libraries_one_mile_buffer <-
  portland_libraries |>
  st_buffer(1609.34)

pps_elementary_schools <-
  read_sf("https://raw.githubusercontent.com/rfortherestofus/twenty-five-schools/master/pps_elementary_schools.shp")

pps_elementary_schools_near_libraries <-
  pps_elementary_schools |>
  st_join(portland_libraries_one_mile_buffer) |>
  mutate(has_nearby_library = case_when(
    is.na(library) ~ "Not within one mile of library",
    .default = "Within one mile of library"
  )) |>
  select(school, has_nearby_library)

maplibre(bounds = portland_libraries_one_mile_buffer) |>
  add_fill_layer(
    source = portland_libraries_one_mile_buffer,
    fill_color = "#7570b3",
    fill_opacity = 0.5,
    tooltip = "library",
    id = "portland_libraries"
  ) |>
  add_circle_layer(
    source = pps_elementary_schools_near_libraries,
    circle_color = match_expr(
      "has_nearby_library",
      values = c(
        "Within one mile of library",
        "Not within one mile of library"
      ),
      stops = c(
        "#1b9e77",
        "#d95f02"
      )
    ),
    tooltip = "school",
    id = "schools"
  ) |>

```

```

, |>
add_categorical_legend(
  values = c(
    "Within one mile of library",
    "Not within one mile of library"
  ),
  legend_title = NULL,
  colors = c("#1b9e77", "#d95f02"),
  circular_patches = TRUE
)

```

Resources

- [Chapter 7 of Kyle Walker's book](#) covers spatial analysis with US census data
- Check out documentation on [st_join\(\)](#), [st_filter\(\)](#), [st_combine\(\)](#) and more functions used for geospatial analysis

Make Interactive Maps

Sample Code

```

maplibre(bounds = pps_elementary_schools_near_libraries) |>
  add_fill_layer(
    source = portland_libraries_one_mile_buffer,
    fill_color = "#7570b3",
    fill_opacity = 0.5,
    tooltip = "library",
    id = "portland_libraries"
  ) |>
  add_circle_layer(
    source = pps_elementary_schools_near_libraries,
    circle_color = match_expr(
      "has_nearby_library",
      values = c(
        "Within one mile of library",
        "Not within one mile of library"
      ),
      stroke = c(

```

```

      stops = c(
        "#1b9e77",
        "#d95f02"
      )
    ),
    tooltip = "school",
    id = "schools"
  ) |>
  add_categorical_legend(
    values = c(
      "Within one mile of library",
      "Not within one mile of library"
    ),
    legend_title = NULL,
    colors = c(
      "#1b9e77",
      "#d95f02"
    ),
    circular_patches = TRUE
  )

```

Resources

- Check out [{mapgl}](#) package docs

Report in New Ways with Quarto

Make Many Different Outputs with Quarto

Resources

- Go to the [Quarto](#) website to learn how to make [html](#), [revealjs](#), and [website](#) outputs

Keep Your Quarto Outputs on Brand

Sample Code

```
meta:
  name: R for the Rest of Us
  link: https://rfortherestofus.com

color:
  foreground: "#404e6b"
  primary: "#6cabdd"

typography:
  fonts:
    - family: Inter
      source: google
    - family: IBM Plex Mono
      source: google
  base: Inter
  headings: Inter
  monospace: IBM Plex Mono
```

Resources

- Check out [brand.yml documentation](#).
- Watch my recent video interview with Garrick Aden-Buie, covering consistent branding with brand.yml:

Consistent Branding with brand.yml: Garrick Aden-Buie on Unifying Styling A...



Publish Your Quarto Documents Online

Resources

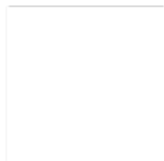
In this [Going Deeper with R course](#) video, I teach how to publish your documents using Netlify. Check it out:

Authorization Granted

i You've granted access to an application named **Quarto**.

Access can be revoked at any time from the [applications screen](#).

You can close this window now and return to the terminal, or you can [browse your sites](#).



Transcript

Click on the transcript to go to that point in the video. Please note that transcripts are auto generated and may contain minor inaccuracies.

0:03 If you create a website or a book with Quarto, you may be wondering, Great, but how do I share this with people? Fortunately, it's actually not very complicated to get your work online. Let me show you how. To publish your work, you're going to run the command Quarto Publish in the terminal. Note that the terminal is separate from the console. Let

Make PDFs with Typst

Sample Code

The best place to see sample code is in [this repo](#), which accompanies my 2024 posit::conf talk (see below).

Resources

My posit::conf(2024) talk covered how to use typst.

David Keyes - Report Design in R: Small Tweaks that Make a Big Difference



Automate all the Things

Email Your Reports Directly from R

Sample Code

```
library(tidyverse)
library(quarto)
library(gmailr)
```

```

rendered_report <- str_glue("covid-business-relief-contact-log-{today()}

quarto_render(
  input = "report.qmd",
  output_file = rendered_report
)

gm_auth_configure()
gm_auth(email = TRUE, cache = ".secret")

email_report <-
  gm_mime() |>
  gm_to("Joe Schmoe <joeschmoe@prosperportland.us>") |>
  gm_from("David Keyes <david@rfortherestofus.com>") |>
  gm_subject("COVID Business Relief Contact Log") |>
  gm_text_body("See attached") |>
  gm_attach_file(rendered_report)

gm_send_message(email_report)

```

Resources

- Check out the [{gmailr} package docs](#)
- Another alternative for sending emails directly from R is the [{blastula} package](#)

Run Your Code Without Lifting a Finger

Sample Code

```

name: Render Report and Send It

on:
  schedule:
    - cron: '00 14 * * 1-5'

jobs:

```



```

build:
  runs-on: ubuntu-latest
  container: rocker/geospatial

  env:
    GMAILR_APP: ${ secrets.GMAILR_APP }
    GMAILR_EMAIL: ${ secrets.GMAILR_EMAIL }

  steps:
    - name: Checkout Repository
      uses: actions/checkout@v2

    - name: Install dependencies
      run: |
        install.packages("remotes")
        remotes::install_cran("quarto")
        remotes::install_cran("gmailr")
        ETC
      shell: Rscript {0}

    - name: Render + Send
      run: |-
        Rscript render_gmail.R

```

Resources

- I demonstrate GitHub Actions in this [blog post and video tutorial](#)
- Albert Rapp has also written a blog post about [automation with GitHub actions](#)

Use R to Work With Files Created in R

Sample Code

```

library(fs)
library(zip)
library(tidyverse)
library(googledrive)

```

```
county_pages <- dir_ls("outputs/pages/county/")

measure_pages <- dir_ls("outputs/pages/measure/")

all_pages_zip <-
  zip(
    zipfile = str_glue("outputs/zip/obtn-files-{today()}.zip"),
    files = c(county_pages, measure_pages)
  )

drive_upload(all_pages_zip)
```

Resources

See documentation for the [{fs}](#), [{zip}](#) and [{googledrive}](#) packages.

[We also have a blog post on working with {fs}.](#)

Use AI to Write Better Code

Create Custom Instructions

Sample Code

My custom instructions are: "Please answer the following R question. When I program, I always like to use the tidyverse. Please don't ever give me base R solutions. Please also always use the native pipe (`|>`) not the tidyverse pipe (`|>`)."

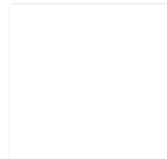
Use AI Directly in your Code Editor

Resources

In the Using AI with R course, I teach how to use AI in both RStudio and Positron.

Using AI in RStudio:

2016	On the job	It was required for my work
2009	By myself using a book or online documentation that ...	It was part of a course I took
2014	Online course (e.g., Coursera, edX, Datacamp)	I thought it would open new career opport
2010	By myself using a book or online documentation that ...	I was personally interested
2018	By myself using a book or online documentation that ...	I thought it would open new career opport
2012	On the job	I wanted to do statistical analyses
2010	By myself using a book or online documentation that ...	I was personally interested
2012	By myself using a book or online documentation that ...	I wanted to do statistical analyses
2010	By myself using a book or online documentation that ...	I wanted to do statistical analyses
2017	By myself using a book or online documentation that ...	I thought it would open new career opport
2012	In-person course (including bootcamps and worksho...	I wanted to do statistical analyses
2016	By myself using a book or online documentation that ...	I wanted to use a specific R package (for e

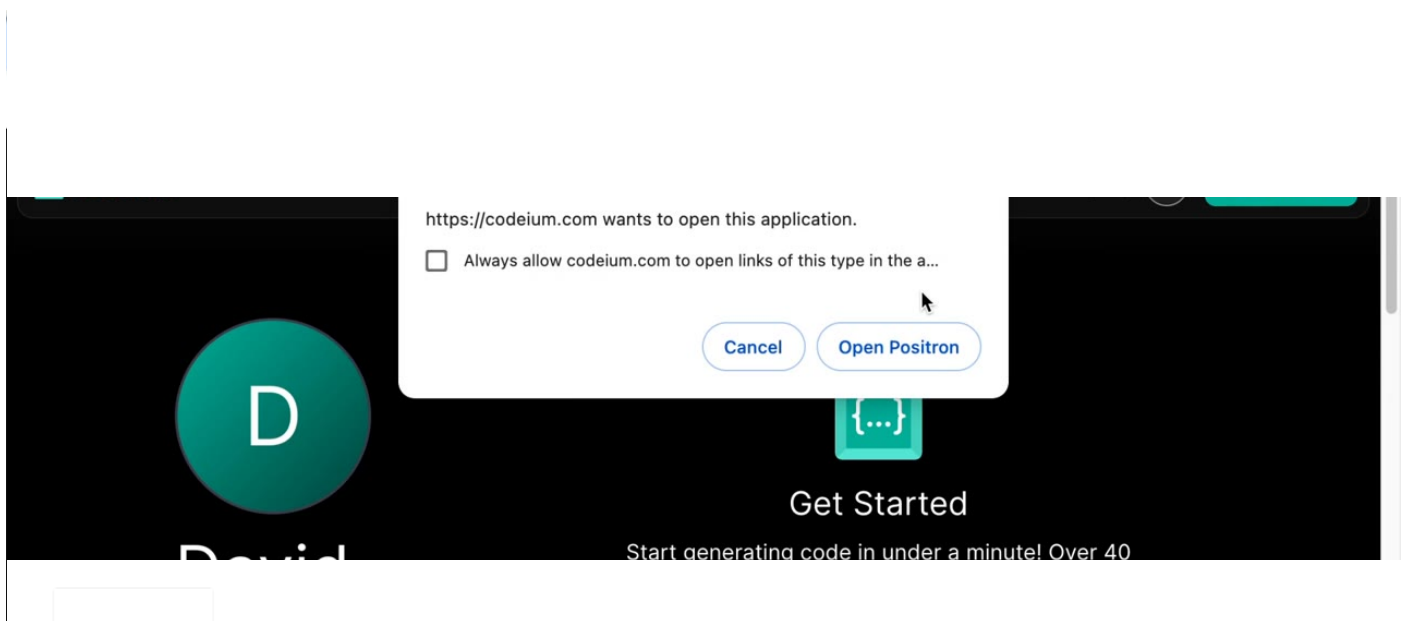


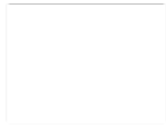
Transcript

Click on the transcript to go to that point in the video. Please note that transcripts are auto generated and may contain minor inaccuracies.

0:03 There are currently two main editors that people use when working in R. The first is RStudio. It's the most common editor and it's been around for a really long time. There's also Positron. This is a new editor that was released in 2024. I'll show you how to work with AI in both of them. But in this lesson, we'll start working in RStudio. In this lesson,

Using AI in Positron:





Transcript

Click on the transcript to go to that point in the video. Please note that transcripts are auto generated and may contain minor inaccuracies.

0:03 Next I want to show how you can use AI in the positron editor if You've never used positron here's what it looks like you've got the explorer on the left side this shows all of your files in the middle you've got the active file that you're working with and then on the right side you've got variables if you make any plots will show up in the bottom right here

Show AI Your Data

Resources

Simon Couch shows how the [{gander}_package](#) can make tedious coding tasks faster and more efficient in a recent R for the Rest of Us podcast interview.

Automating R Workflows with AI



Use AI for Data Analysis

Translate Text

Sample Code

```
library(tidyverse)
library(mall)

survey_spanish <-
  read_csv("https://raw.githubusercontent.com/rfortherestofus/twenty-fi

survey_translated <-
  survey_spanish |>
  llm_translate(spanish, language = "English", pred_name = "english")
```

Resources

Check out the [{mall} package docs](#)

Summarize Text

Sample Code

```
survey_translated_summary <-
  survey_translated |>
  llm_summarize(english, max_words = 5, pred_name = "summary")
```

Resources

Learn about the [llm_summarize\(\) function](#)

Create your Own Prompt to Analyze Text

Sample Code

```
library(ellmer)

identify_themes <- function(text) {
  chat <- chat_openai(
    system_prompt = "You are a sociologist,
    looking for the top three themes in the responses to a survey.
    Each response is separated by \n"
  )

  chat$chat(text)
}

survey_translated_combined <-
  survey_translated |>
  pull(english) |>
  paste(collapse = "\n")

survey_translated_combined |>
  identify_themes()
```

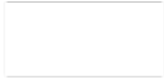
Resources

I show how to use {ellmer} in my [Using AI with R course](#):

Using AI in RStudio

Search

	2014	Online course (e.g., Coursera, edX, Datacamp)	I thought it would open new career opport
	2010	By myself using a book or online documentation that ...	I was personally interested
	2018	By myself using a book or online documentation that ...	I thought it would open new career opport
	2012	On the job	I wanted to do statistical analyses
	2010	By myself using a book or online documentation that ...	I was personally interested
	2012	By myself using a book or online documentation that ...	I wanted to do statistical analyses
	2010	By myself using a book or online documentation that ...	I wanted to do statistical analyses
	2017	By myself using a book or online documentation that ...	I thought it would open new career opport



Check out the [{ellmer} package docs](#)

Join the Community

I wrote a blog post a long time ago about [how the R community became so welcoming](#).

Learn More about R for the Rest of Us

If you're interested in learning more R, check out our [courses](#). And if you'd like to hire us to help you communicate more effectively and efficiently, check out our [consulting services](#).

Sign up for the newsletter

Get blog posts like this delivered straight to your inbox.

Enter your email

Subscribe

Let us know what you think by adding a comment below.

You need to be signed-in to comment on this post. [Login](#).



By David Keyes

June 20, 2025



Sign up for the newsletter

R tips and tricks straight to your inbox.

Enter your email

Subscribe

Search...



R tips and tricks straight to your inbox.

Enter your email

Subscribe

Topics Courses

Using AI with R
Making Beautiful Tables with R
Mapping with R
The Glamour of Graphics
Using Git and GitHub with R
Data Cleaning with R
Package Development with R
Inferential Statistics with R
Interactive Dashboards with Shiny

Core Courses

Getting Started With R
Fundamentals of R
Going Deeper with R

Cohort Courses

R in 3 Months

Other Ways to Learn

Blog
Book
Resources
Podcast
Newsletter
What's New in R

For Organizations

Training
Consulting

About

About
Contact

